

CE 3111

STRUCTURAL ANALYSIS AND DESIGN - I

SECTION A CLASS NOTE

SET 1-4 | Concept + Formula + Shortcut + Correct Sketch + Question Bank Match

SET 1	Stability & Determinacy + SFD/BMD
SET 2	Truss + Three-Hinged Arch
SET 3	Wind + Earthquake
SET 4	Unit Load / Virtual Work + Conjugate Beam

NOTE SCOPE This note follows the Section A = Set 1-4 convention used in the previously prepared Question Bank Analysis. Historical papers sometimes print the two section blocks in reverse order; identify the set by topic, not only by the old printed letter.

Source basis: uploaded CE 3111 lecture slides, Kabbo class notes, QS analysis, 2016-2024 QB, 2024 Backlog and topic-wise solved QB files.

Made by 2223033

How to Use This Note

তোমাকে Section A থেকে 4 সেটের মধ্যে 3 সেট উত্তর করতে হবে। তাই নোটটি চারটি full set ধরে সাজানো হয়েছে। প্রতিটি set-এর ভিতরে theory + numerical workflow + memory shortcut + QB match রাখা হয়েছে।

Set	Core topics	Exam nature	Suggested role
1	Stability/Determinacy + SFD/BMD	Classification + diagram	Backup/Core depending on strength
2	Plane Truss + Three-Hinged Arch	Member force + arch N/Q/M	Core
3	Wind + Earthquake	Table-driven algorithm	Core; highly procedural
4	Unit Load + Conjugate Beam	Deflection + repeated theory/proof	Core

3-of-4 STRATEGY If time is limited: make Set 2, Set 3 and Set 4 exam-ready first; keep Set 1 as a strong backup. If SFD/BMD is your strongest area, swap Set 1 with the set you find harder.

Quick Memory Map

Topic	Memory shortcut
Supports	Roller-Pin-Fixed = 1-2-3 reactions
SFD/BMD	Load $\rightarrow$ Shear $\rightarrow$ Moment; derivative chain: $w \rightarrow V \rightarrow M$
Truss	Joint first if ≤ 2 unknowns; section cut ≤ 3 unknown member forces
Arch	Y-H-M-NQ: ordinate $y \rightarrow$ thrust $H \rightarrow$ moment $M \rightarrow$ resolve N,Q
Wind	$q \rightarrow p \rightarrow F$: sustained pressure $\rightarrow$ design pressure $\rightarrow$ floor force
Earthquake	$T \rightarrow C_s \rightarrow S_a \rightarrow V \rightarrow F_x \rightarrow V_x$
Unit Load	Real M + Virtual $m \rightarrow$ multiply $\rightarrow /EI \rightarrow$ integrate
Conjugate Beam	M/EI is load; $V^* = \theta$, $M^* = \Delta$

Contents

SET 1 - Stability & Determinacy + SFD/BMD

SET 2 - Plane Truss + Three-Hinged Arch

SET 3 - Wind + Earthquake

SET 4 - Unit Load / Virtual Work + Conjugate Beam

Final Formula Sheet + 3-of-4 Checklist

SET 1 - Stability & Determinacy + SFD/BMD

QB STATUS Annual-core family in the complete regular QB reviewed; 2024 Backlog also confirms the same pattern. Usual combination: classification/theory + complete frame/beam force diagrams.

1.1 Structural Idealization and Static Equilibrium

Structural analysis-এর প্রথম কাজ হলো বাস্তব structure-কে এমন idealized model-এ রূপ দেওয়া যাতে support, joint, member এবং load পরিকারভাবে দেখা যায়। Equilibrium মানে structure-এর মোট force এবং moment balance করা।

2D equilibrium equations

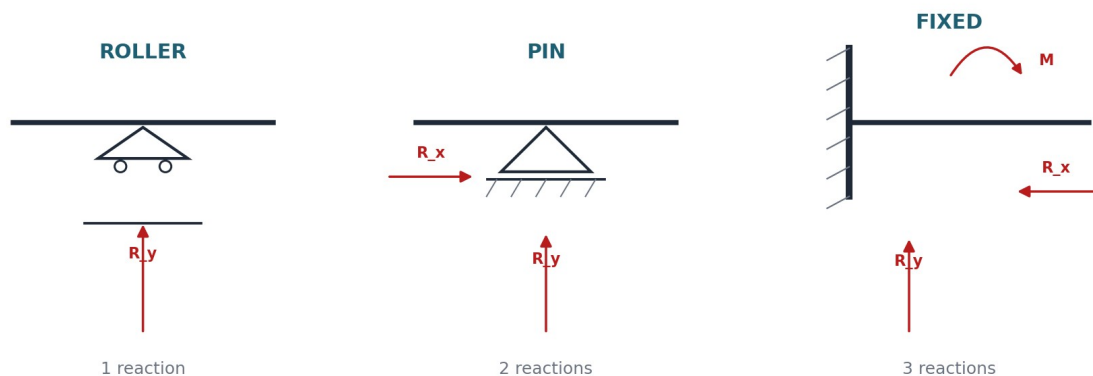
$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M = 0$$

বাংলায় মনে রাখো Horizontal force, vertical force এবং moment - এই তিন balance হলেই 2D rigid body equilibrium পাওয়া যায়।

1.2 Supports and Reaction Components



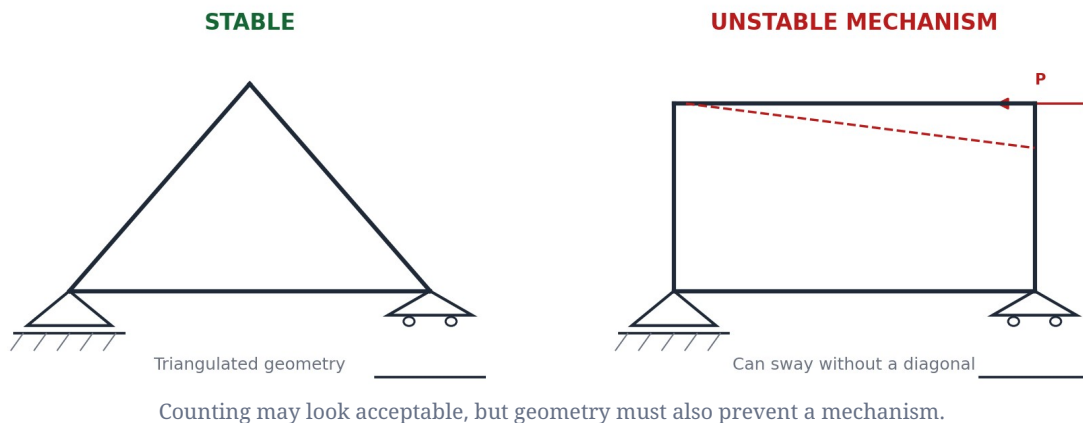
Memory trick: Roller = 1, Pin = 2, Fixed = 3 reaction components.

Support / Connection	What it restrains	Reaction unknowns	Exam note
Roller	Translation normal to supporting surface	1	Rotation and tangential movement are allowed.
Pin / Hinge support	Two translations	2 (R_x , R_y)	Rotation is allowed; no support moment.
Fixed support	Two translations + rotation	3 (R_x , R_y , M)	Support moment generally exists.
Internal hinge	Relative rotation at joint; moment release	2 internal force components when structure is split	Bending moment at the hinge = 0.

Shortcut External support count is only a first check. Internal hinges and geometry can change the number of rigid bodies and can create mechanisms.

1.3 Determinate, Indeterminate, Stable and Unstable

Term	Exam-ready meaning
Statically determinate	All required reactions/internal forces can be found from equilibrium equations only.
Statically indeterminate	Unknown force components exceed independent equilibrium equations; compatibility/deformation relations are also required.
Stable	Supports and geometry prevent rigid-body motion/mechanism under small disturbance.
Unstable	Structure can move as a rigid body or mechanism; such a structure is unacceptable regardless of force count.



Counting idea for a 2D assembly

যদি internal hinge দিয়ে structure-কে কয়েকটি rigid body-তে ভাগ করা যায়, প্রতিটি rigid body-এর জন্য 3টি independent equilibrium equation থাকে। তাই n টি rigid body হলে মোট $3n$ equation পাওয়া যায়। Split করার সময় internal hinge force components-ও unknown হিসেবে ধরতে হবে।

Rigid-body counting idea used in the lecture examples

$$\text{Available independent equilibrium equations} = 3n$$

Critical caution Do not conclude “stable” only because unknowns = equations. Parallel/concurrent reaction lines or collapsible geometry may still make a mechanism.

Classification checklist - 20 second exam method

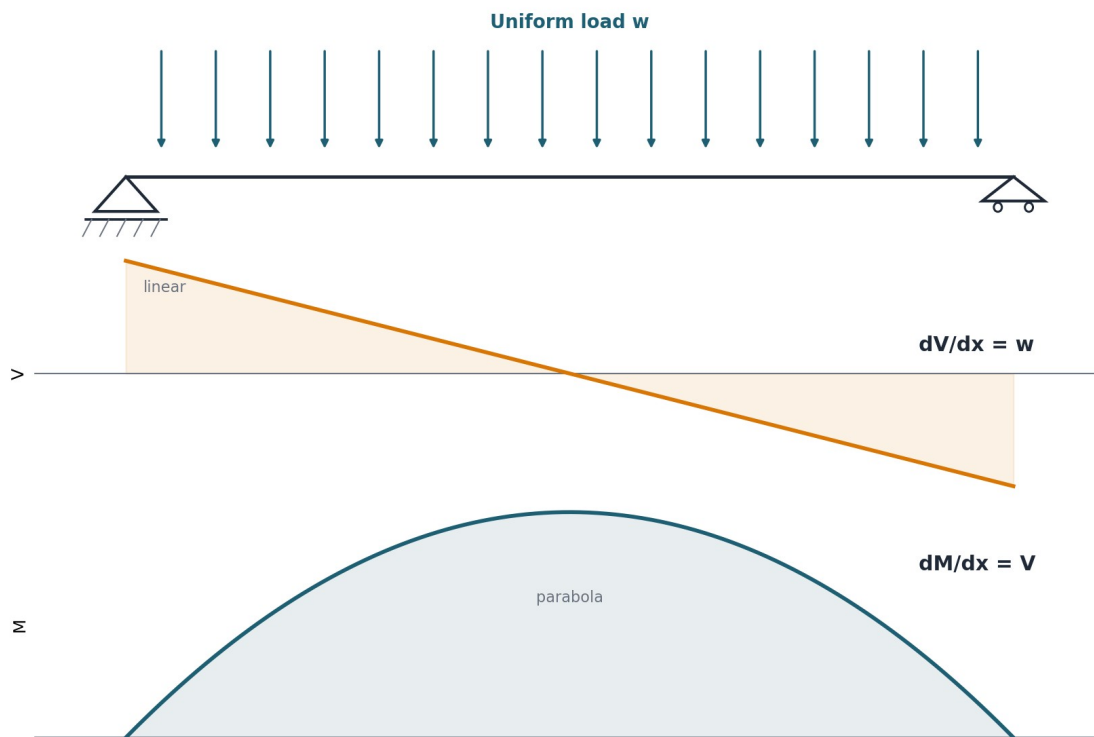
1. Identify every support and its reaction direction.
2. Locate internal hinges/releases; remember $M = 0$ at a hinge.
3. Count rigid bodies/components after splitting at hinges.
4. Compare unknown force components with available equilibrium equations.
5. Finally inspect geometry: can it translate/rotate/sway as a mechanism? If yes $\rightarrow$ unstable.

1.4 Internal Forces: N, V and M

A cut section of a coplanar member generally carries axial force N , shear force V and bending moment M . The lecture sign convention treats tensile axial force as positive; positive bending produces compression at the top and tension at the bottom (sagging-type positive moment).

Quantity	Physical meaning	Fast check
N - Axial force	Along member axis	Tension stretches; compression shortens.
V - Shear force	Transverse resultant	Point load causes a jump in SFD.
M - Bending moment	Internal couple	Applied couple causes a jump in BMD; hinge has $M=0$.

1.5 SFD/BMD Core Relations



Lecture relation: $dV/dx = w$ and $dM/dx = V$ for the stated sign convention.

Load - shear - moment relations

$$dV/dx = w$$

$$dM/dx = V$$

বাংলা **shortcut** Load diagram-এর area $\rightarrow$ shear change; shear diagram-এর area $\rightarrow$ moment change. অর্থাৎ “Area নিচে গেলে পরের diagram-এর ordinate বদলায়”.

Event on member	SFD response	BMD response
Concentrated vertical force	Sudden jump by force magnitude	Moment remains continuous if no applied couple
Concentrated couple	No direct jump in shear	Sudden jump by couple magnitude
UDL	Linear variation	Parabolic variation
No distributed load	Constant shear	Linear moment
Internal hinge	Shear may be nonzero	$M = 0$ at hinge
Location where $V = 0$	-	Possible local max/min of M

1.6 Beam / Frame Solution Workflow

6. Draw the complete FBD and solve support reactions first.
7. For frames, isolate members/joints when necessary and use equal-and-opposite joint actions.
8. Start the SFD from a convenient free end/support and move member-by-member.
9. Use equilibrium or area relation to obtain moment ordinates.
10. At an internal hinge enforce $M = 0$; at a free end $M = 0$ unless an external couple is applied.
11. Label sign and units on every diagram. Check that end/joint moments satisfy joint equilibrium.

Frame exam technique Do not “draw by shape only”. First write the end force/end moment values, then connect them with the correct constant/linear/parabolic form.

1.7 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Classify structures; draw SFD/BMD of a loaded frame.
2017	Differentiate determinate/indeterminate and stable/unstable; classify structures; frame force diagrams.
2018	Zero-force-member purpose + classification; reactions and SFD/BMD of frame.
2019	Classification + frame SFD/BMD with given end moments.
2020*	Classification visible in shortened scan.
2021	Classification; overhang beam SFD/BMD appears in the same core block.
2022	Zero-force member + classification; frame reaction/SFD/BMD.
2023	Determinacy/stability classification; pitched-frame force diagrams.
2024	Classification + complete frame force-diagram problem.
2024 BL	Differentiate + classify; reactions and axial/shear/bending diagrams of frame.

Set 1 - Common Errors

- Counting reactions but ignoring geometry/mechanism.
- Giving a moment reaction at a pin/roller.
- Forgetting $M = 0$ at an internal hinge.
- Mixing member sign conventions midway through a frame solution.
- Drawing a curved BMD under a region where V is constant.

SET 2 - Plane Truss + Three-Hinged Arch

QB STATUS Truss member force + three-hinged arch point-force/moment problems recur almost every year. Recent papers also attach arch theory or zero-force concepts.

2A. Plane Truss

2A.1 Definition and Idealization

Truss হলো slender straight members-এর pin-connected arrangement যেখানে ideal analysis-এ load joint-এ প্রয়োগ করা হয় এবং প্রতিটি member মূলত axial force বহন করে।

Assumption	Meaning in calculation
Frictionless pin joints	No joint moment is transferred in the ideal model.
Loads act at joints	Members are treated as two-force members.
Member self-weight neglected or joint-loaded	Avoids distributed bending in individual members.
Centroidal member axes meet at joint	Prevents eccentric moment in the idealization.

2A.2 Determinacy and Stability of a Plane Truss

Plane-truss count

$$b + r = 2j \rightarrow \text{statically determinate (if stable)}$$

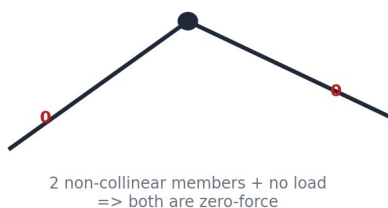
$$b + r > 2j \rightarrow \text{statically indeterminate}$$

$$b + r < 2j \rightarrow \text{unstable}$$

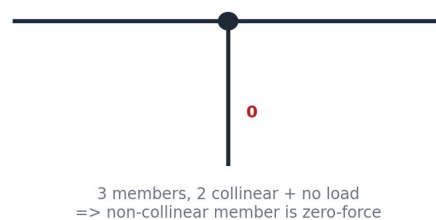
where b = number of bars, r = external support reaction components, j = joints. এই counting condition পূরণ হলেও reaction lines concurrent/parallel হলে বা internal geometry collapsible হলে truss unstable হতে পারে।

2A.3 Zero-Force Members

RULE 1



RULE 2



Zero-force members are identified by inspection before long calculations.

Why they exist Zero-force member load-carrying condition বদলালে useful হতে পারে, truss shape/stability বজায় রাখতে পারে, এবং future/reversed loading-এর জন্য path দেয়. QB often asks purpose + identification.

2A.4 Method of Joints

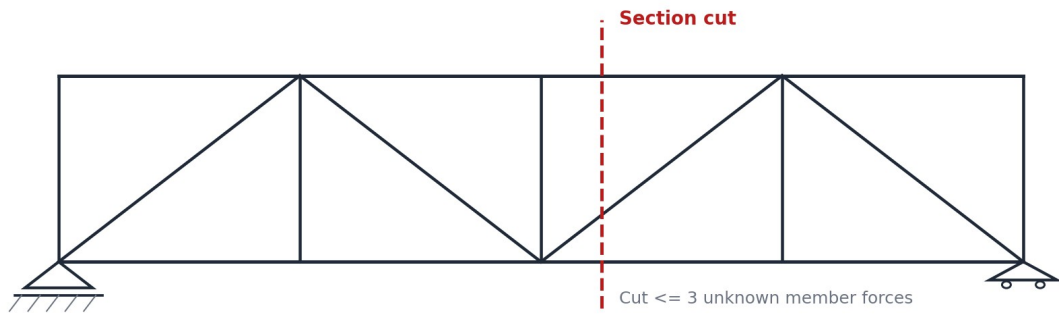
12. Find external support reactions from the whole-truss FBD.
13. Start at a joint with at most two unknown member forces whenever possible.
14. Assume unknown member forces in tension (arrows away from the joint).
15. Apply $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

16. Move to the next joint where only one/two unknowns remain.

17. Positive under the assumed tension direction $\rightarrow$ T; negative $\rightarrow$ actual compression C.

Memory shortcut "2 unknown joint first." If you start at a joint with 3 unknown member forces, two 2D equations are not enough.

2A.5 Method of Sections



Choose the cut to pass through the required members and not more than three unknown member forces.

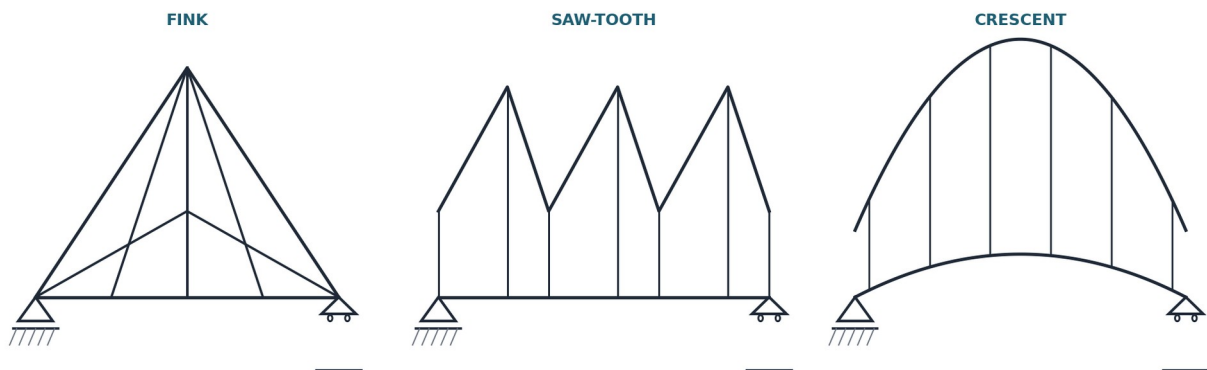
18. Cut the truss into two parts through the desired members (normally ≤ 3 unknown member forces).

19. Choose the easier side of the cut.

20. Draw its FBD and assume cut-member forces in tension.

21. Use $\Sigma M = 0$ strategically to eliminate two unknowns at once; then use ΣF_x and ΣF_y .

2A.6 Typical Truss Shapes Seen in QB



Fink, saw-tooth and crescent truss sketches - useful for short theory/identification.

2A.7 Truss QA Checklist

- Reaction first, member force later.
- Angles must come from geometry, not visual guess.
- Write T/C beside every final member force.
- Check one solved joint with both ΣF_x and ΣF_y before moving on.

2B. Three-Hinged Arch

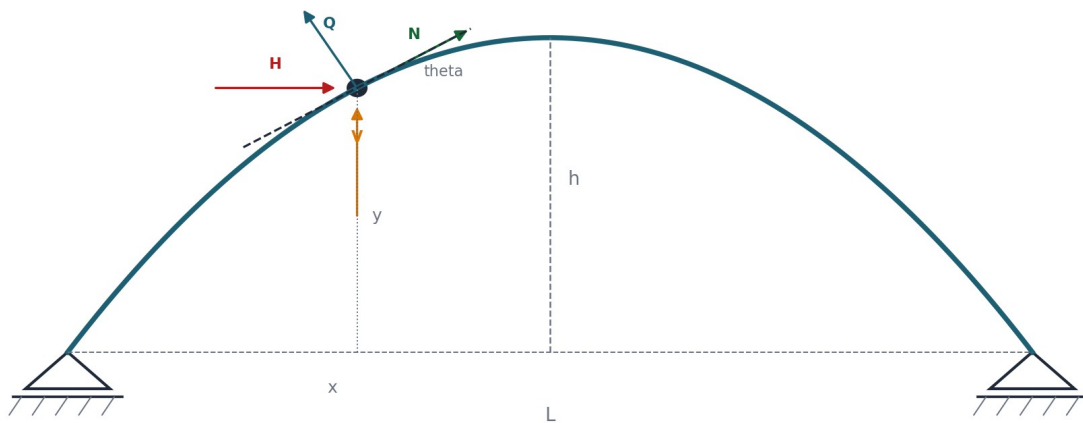
2B.1 Why an Arch?

Arch একটি curved girder. Curvature-এর কারণে horizontal reaction তৈরি হয়, যা equivalent beam-এর bending moment কমায়। তাই long span-এ arch transverse load efficiently বহন করতে পারে এবং অনেক load axial compression হিসেবে যায়।

Arch type	Static character	This course focus
Three-hinged arch	Statically determinate	YES - reactions and internal forces from equilibrium
Two-hinged arch	Statically indeterminate	Concept only in this lecture block
Fixed arch	Statically indeterminate	Concept only in this lecture block

2B.2 Parabolic Arch Geometry

Three-hinged parabolic arch: geometry + local forces



At section P(x,y), resolve global H and vertical resultant V into local normal N and shear Q.

Three-hinged parabolic arch

$$y = 4 h x (L - x) / L^2$$

$$dy/dx = 4 h (L - 2x) / L^2 = \tan(\theta)$$

Memory shortcut First find y. Without the correct ordinate y, horizontal thrust H cannot be converted into the correct bending-moment reduction $H \cdot y$.

2B.3 Reaction and Horizontal Thrust Workflow

22. Use whole-arch equilibrium to find vertical reactions R_A and R_B (and any unbalanced horizontal load effect).
23. Use moment = 0 at the crown hinge C to obtain horizontal thrust H.
24. For a chosen section, calculate y from the parabolic equation and determine the net vertical resultant V on the chosen side.
25. Find arch bending moment from ordinary beam moment minus $H \cdot y$.
26. If N and Q are asked, calculate tangent angle θ and resolve H,V into local axes.

Local force resolution (with V taken algebraically upward on the chosen free body)

$$M_{\text{arch}} = M_{\text{beam}} \text{ at same section} - H y$$

$$N = H \cos(\theta) + V \sin(\theta)$$

$$Q = V \cos(\theta) - H \sin(\theta)$$

Sign note N and Q signs depend on the local-axis convention. The equations above match the resolution convention used in the uploaded solved arch notes when V is the algebraic vertical resultant on the selected side.

2B.4 Maximum Bending Moment in an Arch

Write $M(x)$ correctly first. Then set $dM/dx = 0$ to locate a local maximum/minimum and substitute that x back into $M(x)$. Do not differentiate before including the $H \cdot y$ term.

Extremum condition

$$dM(x)/dx = 0 \rightarrow \text{candidate location of maximum/minimum bending moment}$$

2B.5 Fast Arch Sequence

Y-H-M-NQ Y = section ordinate $\rightarrow$ H = horizontal thrust from crown-hinge equilibrium $\rightarrow$ M = beam moment - $H \cdot y \rightarrow$ N, Q = resolve with theta. This order prevents most arch mistakes.

2B.6 Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Selected truss members by method of joints; arch bending moment at a specified point.
2017	Selected truss member forces; three-hinged arch internal forces/moment at D.
2018	Method of sections for selected members; arch BM at specified distance.
2019	Henneberg/typical truss shapes; arch BM at specified distance.
2020*	Arch BM at a specified point is visible.
2021	Cantilever-truss/limiting-load truss; arch pure-compression condition.
2022	Method of joints; arch advantages and maximum BM location/magnitude.
2023	All-member truss under UDL; beam-vs-arch comparison + internal forces.
2024	Selected truss member forces; three-hinged arch internal force/BM.
2024 BL	FG/GD/CB/CF truss forces; arch internal forces and moment at D.

Set 2 - Common Errors

- Using $b+r=2j$ and forgetting geometric stability.
- Starting method of joints at a joint with too many unknowns.
- Not writing T/C in the final truss answer.
- Using span coordinate x from the wrong support in arch equation.
- Using M_{beam} only and forgetting the $H \cdot y$ reduction.
- Resolving N/Q before calculating the correct tangent angle.

SET 3 - Wind + Earthquake

QB STATUS This is the most algorithmic Section A set. The values and coefficient tables change from paper to paper, but the calculation sequence is highly repetitive.

3A. Wind Load Analysis

3A.1 Core Concepts

Wind হলো moving air এবং building-এর জন্য এটি lateral load. Design wind action depends on wind speed, building size/shape, terrain exposure, surrounding site profile and adjacent structures.

Term	Source-based meaning
Basic wind speed V_b	Fastest-mile wind speed in km/h referenced at 10 m above ground terrain in the lecture source.
Sustained wind pressure q_z	Computed incident pressure of a uniform air stream at height z for a terrain/exposure and occupancy condition.
Design wind pressure p_z	Equivalent static pressure including gust effect, used for building design.
Terrain / Topography	Surface roughness / major landform features that alter wind flow.
Importance factor C_I	Accounts for hazard to life/property according to occupancy/importance.

3A.2 Types / Characteristics That Can Become Theory Questions

- Prevailing wind, seasonal wind and local wind.
- Variation of wind velocity with height.
- Wind turbulence.
- Statistical probability / return-period character.
- Vortex shedding phenomenon.
- Dynamic wind-structure interaction.

Vortex shedding - Bangla explanation Bluff body-এর দুই পাশে alternating vortex ছুটে গেলে cross-wind oscillation তৈরি হতে পারে. যদি shedding frequency structural natural frequency-এর কাছাকাছি হয়, vibration গুরুত্বপূর্ণ হয়ে যায়.

3A.3 Wind Pressure Formula Chain

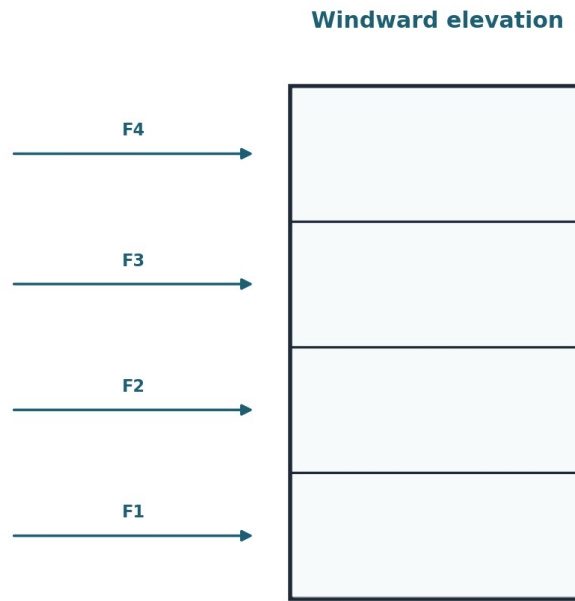
Lecture equations for sustained and design wind pressure

$$q_z = C_I C_C C_z V_b^2$$

$$p_z = C_G C_p q_z$$

Symbol	Meaning
q_z	Sustained wind pressure at height z (kN/m ²)
C_I	Structure importance coefficient
C_C	Velocity-to-pressure conversion coefficient (lecture gives 47.2×10^{-6})
C_z	Combined height and exposure coefficient
V_b	Basic wind speed (km/h)
C_G	Gust coefficient / gust response factor as specified
C_p	Pressure coefficient
p_z	Design wind pressure at height z

3A.4 Floor-wise Wind Load Workflow



$$\text{Floor force} = \text{design pressure} \times \text{tributary projected area}$$

Use the projected tributary area for each floor strip in the wind direction.

27. Read V_b , exposure category, importance category/coefficient and C_p from the question.
28. At each floor/height, take C_z and C_G (or G_h/G_z as supplied) from the table in that paper.
29. Compute q_z , then p_z .
30. Find each floor tributary projected area from the plan/elevation.
31. Floor wind force = $p_z \times$ tributary projected area.
32. Check units: $\text{kN/m}^2 \times \text{m}^2 = \text{kN}$.

Exam shortcut Write one calculation table with columns: z | C_z | C_G | q_z | p_z | tributary area | floor force. This prevents coefficient mixing.

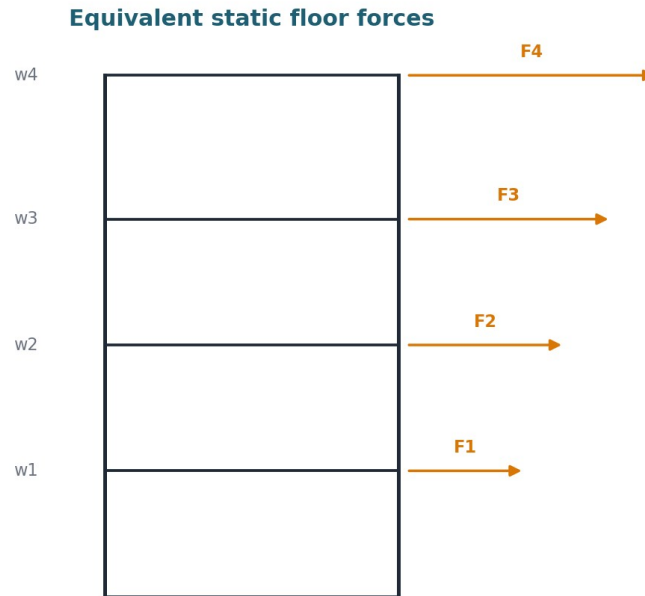
Important Use the coefficient table supplied in the examination problem. Old and recent papers do not always use identical data conventions.

3B. Earthquake / Seismic Load Analysis

3B.1 Essential Definitions

Term	Exam-ready definition
Magnitude	Measure related to energy released by an earthquake.
Intensity	Measure of ground shaking at a particular site.
Base shear	Total design lateral seismic force at the base.
Storey shear	Total horizontal shear at a particular storey from forces at that level and above.
Storey drift	Horizontal displacement of top of a storey relative to its bottom.
Site class	Classification based on upper soil profile/properties (lecture uses upper 30 m).

3B.2 Equivalent Static Force Method - Core Sequence



Higher floor height increases the distribution weight $w \cdot h^k$

Equivalent static method converts seismic action into a set of lateral floor forces.

Main sequence from the earthquake lecture

$$T = C_t (h_n)^m$$

$$S_a = (2/3) (Z I / R) C_s$$

$$V = S_a W$$

$$F_x = V (w_x h_x^k) / \sum (w_i h_i^k)$$

$$V_x = \sum F_i \text{ (from storey } x \text{ to roof)}$$

Step	What to do
1. Period T	Estimate from C_t , building height h_n and exponent m if the problem requires it.
2. Site spectrum C_s	Use site class + period and the supplied/lecture spectrum relation/table.
3. Design spectral acceleration S_a	Combine Z , I , R and C_s using the lecture equation; observe any stated lower bound.
4. Base shear V	Multiply S_a by total seismic weight W .
5. Vertical distribution F_x	Distribute V using $w_i h_i^k$ weighting.
6. Storey shear V_x	Sum floor forces from the considered storey to the roof.

3B.3 Exponent k for Floor-force Distribution

Building period T	k used in lecture
$T \leq 0.5 \text{ s}$	$k = 1$
$T \geq 2.5 \text{ s}$	$k = 2$
$0.5 < T < 2.5 \text{ s}$	Linear interpolation between 1 and 2

3B.4 Site Factor Table - Values Shown in Lecture

Site class	S	TB (s)	TC (s)	TD (s)
SA	1.00	0.15	0.40	2.0
SB	1.20	0.15	0.50	2.0
SC	1.15	0.20	0.60	2.0
SD	1.35	0.20	0.80	2.0
SE	1.40	0.15	0.50	2.0

Do not memorize blindly QB questions often provide the required soil/site parameters directly. Use the values specified in the question first; use lecture tables only when the problem asks for them.

3B.5 Equivalent Static vs Dynamic Response

Method	Idea
Equivalent Static	Simplifies earthquake action into static lateral forces; starts with design base shear.
Response Spectrum	Linear elastic dynamic analysis using modal response and spectral accelerations.
Time History	Numerical integration using ground acceleration time records; may be linear or nonlinear.

3C. Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Sustained/design wind-pressure theory + floor-wise wind; floor-wise earthquake load.
2017	5-storey wind; 6-storey hospital seismic by equivalent static method.
2018	5-storey seismic + 5-storey wind.
2019	4-storey seismic + 4-storey wind.
2020*	4-storey wind + 5-storey seismic visible in shortened scan.
2021	4-storey wind; 5-storey steel-building seismic.
2022	Vortex shedding + 7-storey wind; 5-storey earthquake.
2023	Site-specific 7-storey wind + 5-storey seismic with supplied tables.
2024	Wind + equivalent-static earthquake again.
2024 BL	5-storey wind + 5-storey earthquake full numerical problems.

Set 3 - Common Errors

- Using C_z/G values from another year instead of the table supplied.
- Using total plan area instead of projected tributary area normal to wind direction.
- Forgetting to convert pressure $\times$ area into floor force.
- For seismic distribution, using floor number instead of actual height h_i .
- Forgetting the exponent k or summation denominator.
- Mixing total base shear V with storey shear V_x .

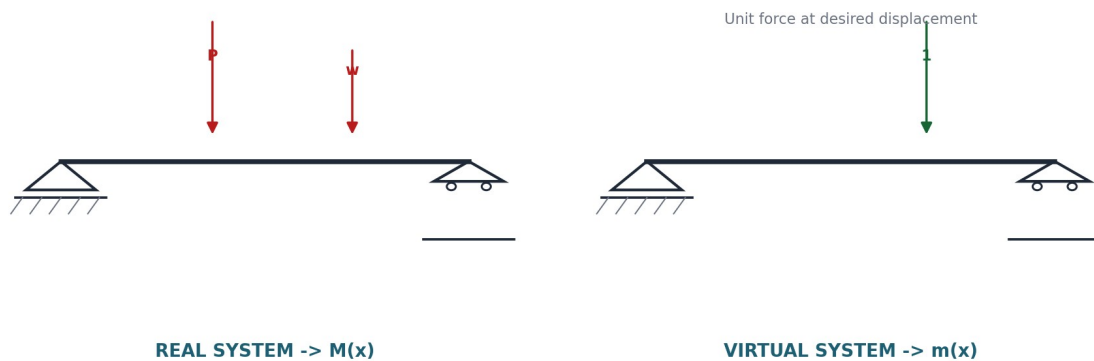
SET 4 - Unit Load / Virtual Work + Conjugate Beam

QB STATUS A highly dependable pair: one unit-load/virtual-work problem and one conjugate-beam theory/numerical. The unit-load deflection proof repeats directly in the QB.

4A. Unit Load / Virtual Work Method

4A.1 Principle

Virtual work method-এ desired displacement-এর দিকেই একটি imaginary unit action বসানো হয়। Real loading যে internal force/moment তৈরি করে এবং unit action যে virtual internal force/moment তৈরি করে - তাদের work product থেকে displacement পাওয়া যায়।



Translation needs a virtual unit force; rotation needs a virtual unit moment/couple.

Core principle

$$\text{Virtual external work} = \text{Virtual internal work}$$

4A.2 Beam and Frame Formula

Bending-dominated beam/frame deflection

$$\Delta = \int (m M / E I) dx$$

$$\theta = \int (m_{\theta} M / E I) dx$$

Symbol	Meaning
M	Bending-moment function due to real loading
m	Bending-moment function due to virtual unit force at target displacement
m _{theta}	Moment function due to virtual unit couple at target rotation
EI	Flexural rigidity; split the integral where E or I changes
Delta / theta	Desired translation / rotation

4A.3 Truss Formula

Axial-deformation form for trusses

$$\Delta = \sum (n N L / A E)$$

N = real member axial force; n = member force under virtual unit load; L, A, E = member length, area and modulus. QB has used unit load on truss joints, so the sign of N and n matters.

4A.4 Unit Load Workflow

33. Solve real reactions and obtain real $M(x)$ for each beam/frame segment (or N for each truss member).
34. Apply a unit force at the target translation, or a unit moment at the target rotation, in the positive desired direction.
35. Solve the virtual system and obtain $m(x) / n$.
36. Split the structure at load changes, hinges, joints, or EI changes.
37. Integrate mM/EI over each segment, or sum nNL/AE for trusses.
38. Algebraic sign of the answer tells whether actual movement agrees with the assumed unit-action direction.

Memory shortcut REAL M + VIRTUAL m $\rightarrow$ Multiply $\rightarrow$ divide by EI $\rightarrow$ integrate. For rotation, replace unit force by unit moment.

4A.5 Direct Proof You Must Know

For a beam element dx , real bending moment M produces rotation $d\theta = (M/EI) dx$. A virtual unit load produces virtual moment m . Virtual internal work = $m d\theta$. Summing over the beam gives the displacement corresponding to the unit load.

Unit-load deflection derivation

$$d\theta = (M/EI) dx$$

$$1 \times \Delta = \int m d\theta = \int (mM/EI) dx$$

$$\text{Therefore, } \Delta = \int (mM/EI) dx$$

Proof writing technique Write the meaning of M , m , EI and dx after the derivation. The QB asks "where symbols have their usual meaning"; defining them makes the proof complete.

4B. Conjugate Beam Method

4B.1 Definition and Two Theorems

Conjugate beam হলো real beam-এর সমান length-এর একটি fictitious beam. Real beam-এর M/EI diagram-কে conjugate beam-এর distributed load হিসেবে দিলে conjugate beam-এর shear এবং moment যথাক্রমে real beam-এর slope এবং deflection দেয়।

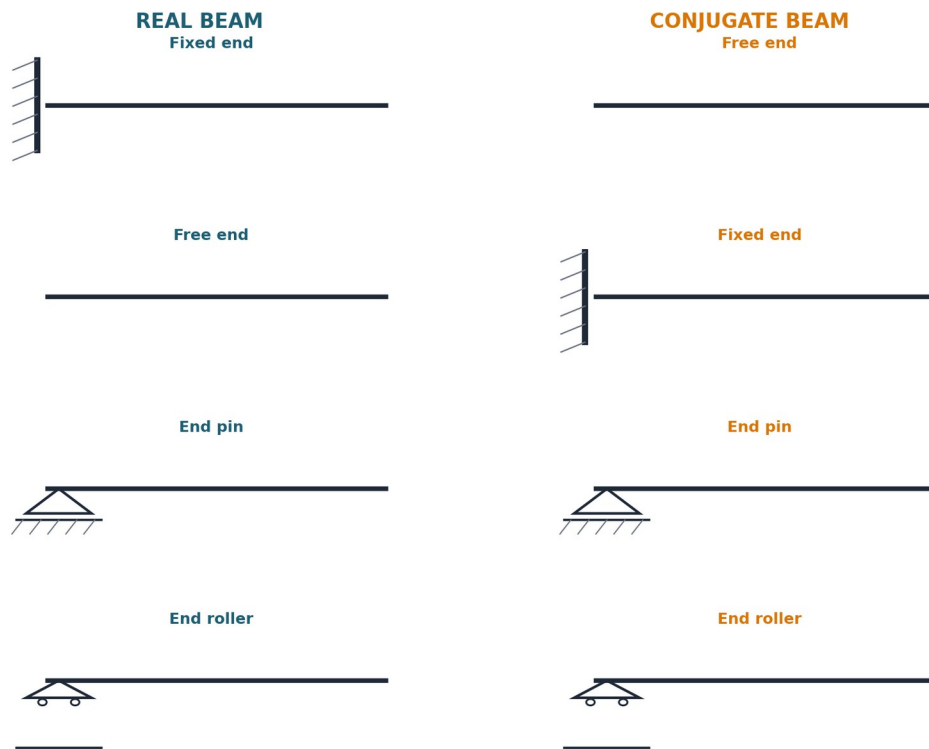
Conjugate-beam theorems

$$V^* \text{ at } x = \theta \text{ at } x \text{ in real beam}$$

$$M^* \text{ at } x = \Delta \text{ at } x \text{ in real beam}$$

Memory shortcut M/EI goes IN; V^* gives θ ; M^* gives Δ . "Load $\rightarrow$ Shear $\rightarrow$ Moment" chain is analogous to "Curvature $\rightarrow$ Slope $\rightarrow$ Deflection".

4B.2 Support Conversion - High Priority



Most-used end-support conversions. The full lecture also treats internal pins/rollers on continuous beams.

Real beam condition	Conjugate beam condition	Reason
Fixed end	Free end	$\theta = 0$ and $\Delta = 0 \rightarrow V^* = 0, M^* = 0$
Free end	Fixed end	θ and Δ are unrestricted $\rightarrow$ conjugate end can develop V^*, M^*
End pin	End pin	$\Delta = 0, \theta$ free $\rightarrow M^* = 0, V^*$ may exist
End roller	End roller	Same displacement/slope logic as an end simple support

Internal support caution Continuous-beam internal hinge/roller conversion is more subtle because slope or conjugate shear may be discontinuous. Follow the lecture support-conversion diagram, not only the four end rules above.

4B.3 Sign Convention

- Positive M/EI ordinate $\rightarrow$ apply upward load on the conjugate beam (as used in the lecture convention).
- Positive conjugate shear $\rightarrow$ positive/counterclockwise real-beam slope.
- Positive conjugate moment $\rightarrow$ positive/upward real-beam deflection.

4B.4 Solution Procedure

39. Solve real-beam reactions and draw the real M diagram.
40. Divide by EI to obtain the M/EI loading; if I changes, scale each region correctly.
41. Convert the real beam to the conjugate beam using support rules.
42. Apply the M/EI diagram as distributed loading with the stated sign.
43. Solve conjugate reactions by ordinary equilibrium.
44. At the target point, conjugate shear = real slope and conjugate bending moment = real deflection.

Fast check If the real beam has zero deflection at a support, the conjugate bending moment at the corresponding point must be zero. If the real beam has zero slope at a fixed end, conjugate shear there must also be zero.

4C. Matching Questions from the Question Bank

Year	Matching question pattern (paraphrased)
2016	Conjugate-beam definition; unit-load frame deflection; multi-output conjugate numerical.
2017	Real-vs-conjugate relation; unit-load frame deflection; conjugate numerical.
2018	Direct proof of unit-load formula; truss-joint deflection by unit load.
2019	Frame rotation/horizontal deflection by virtual work; conjugate moments/max deflection/slope.
2020*	Unit-load deflection proof visible in shortened scan.
2021	Truss horizontal deflection by unit load; conjugate moments/max deflection/slope.
2022	Frame rotation + horizontal deflection; conjugate relation + numerical.
2023	Cantilever slope/deflection by unit load; conjugate slope/deflection at B and D.
2024	Frame joint rotation/deflection; conjugate-beam definition + numerical.
2024 BL	Frame unit-load deflection + conjugate-beam definition/relationship and numerical.

Set 4 - Common Errors

- Applying a unit force when the requested quantity is rotation (must use unit moment).
- Forgetting to split integral when M , m or EI changes.
- Using real M instead of M/EI as conjugate loading.
- Wrong fixed/free support conversion.
- Reading conjugate moment as slope (correct: shear $\rightarrow$ slope, moment $\rightarrow$ deflection).
- Ignoring the sign/direction of the unit action and reporting a negative result without interpretation.

Final Rapid Revision - Section A

One-Page Formula Sheet

Set	Must-know formula / rule
1	2D: $\Sigma F_x=0, \Sigma F_y=0, \Sigma M=0$ Internal hinge: $M=0$ $dV/dx=w, dM/dx=V$
2	Truss: $b+r=2j$ (if stable) Arch: $y=4hx(L-x)/L^2$ $M=M_{beam}-Hy$ $N=H\cos\theta+V\sin\theta$ $Q=V\cos\theta-H\sin\theta$
3	Wind: $q_z=CI CC Cz Vb^2$; $p_z=CG Cp q_z$ Seismic: $T=C_t h_n^m$; $S_a=(2/3)(ZI/R)C_s$; $V=S_a W$; $F_x=V w_x h_x^k / \Sigma w_i h_i^k$
4	Beam/frame: $\Delta = \int m M / EI dx$ Truss: $\Delta = \Sigma n N L / AE$ Conjugate: load= M/EI , $V^*=\theta$, $M^*=\Delta$

Theory Questions - High Priority

Priority	Question
VERY HIGH	Differentiate determinate vs indeterminate and stable vs unstable; classify structures.
HIGH	What is a zero-force member? State rules/purpose.
HIGH	Advantages of arch over beam / difference between beam and arch.
HIGH	Define sustained and design wind pressure; factors governing wind magnitude.
RECENT	Explain vortex shedding effect.
VERY HIGH	Define conjugate beam and state real-beam/conjugate-beam relations.
VERY HIGH	Prove $\Delta = \int m M / EI dx$ by unit-load method.

3-of-4 Decision Checklist

Set	Before choosing it in exam, verify that you can...
Set 1	Classify supports/hinges without mechanism mistakes and draw complete SFD/BMD with signs.
Set 2	Solve truss forces systematically and compute arch y, H, M, N, Q at a section.
Set 3	Finish wind and earthquake calculation tables without mixing coefficients/areas/heights.
Set 4	Set up unit-load mM/EI correctly and convert real supports to conjugate supports correctly.

Last 15-Minute Memory Lines

- Roller-Pin-Fixed = 1-2-3 reactions.
- Internal hinge = zero bending moment.
- Load $\rightarrow$ Shear $\rightarrow$ Moment; area advances you to the next response diagram.
- Truss: start where unknowns ≤ 2 ; section cut ≤ 3 unknowns.
- Arch: Y-H-M-NQ.
- Wind: $q \rightarrow p \rightarrow$ floor force.
- Earthquake: $T \rightarrow C_s \rightarrow S_a \rightarrow V \rightarrow F_x \rightarrow$ storey shear.
- Unit load: translation = unit force, rotation = unit moment.
- Conjugate: M/EI load; $V^* = \theta$; $M^* = \Delta$.

Final warning Historical exam papers may swap printed Section A/B blocks. This note follows the Set 1-4 "Section A" convention used in your Question Bank Analysis. Match by topic on exam day.

Source Basis

- 1.0 - Analysis of Statically Determinate Structures - Stability & Determinacy.pdf
- 2.0 - Shear Force and Bending Moment of Statically Determinate Structures.pdf
- 3.0 - Analysis of Arches.pdf
- 4.0 - Analysis of Wind.pdf
- 5.0 - Earthquake.pdf
- 6.0 - Deflection (Virtual Work Method).pdf
- 7.0 - Deflection (Conjugate Beam Method).pdf
- 0.0 - Classnote - Lec 1-14 - Structure - Kabbo Sir - 2223036.pdf / Structure-Kabbo sir.pdf
- QS ANALYSIS - AYAN SIR + Structural Analysis and Design-I QB (2016-2024) + 2024 Backlog Exam
- Topic-wise solved QB files for Truss and Arch.

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